

# **PSYLIQ — PROJECT 1**

## **Scam Execution & Victim Profiling**

Investigative Data Analysis & Business Insights Report

Analysis Period: 2020–2024

Prepared from the supplied Data\_Matrix\_P1.xlsx

# Executive Summary

This investigation analyzes U.S. scam activity across victim demographics, geographic complaint concentration, scam narratives, urgency language, payment methods, communication platforms, and government enforcement actions.

The supplied age dataset shows the 60+ group rising from 105,301 in 2020 to 147,127 in 2024 (39.7% growth), while ages 20–29 changed from 70,791 to 71,399 (0.9%).

California is the highest-volume state in the supplied 2024 state records. In the narrative sample, “immediately” is the most frequent urgency keyword, cryptocurrency is the most frequent payment method, and phone is the most frequent individual platform after splitting combined platform descriptions.

The government-action table contains 10 actions. Because the supplied actions are concentrated on DOJ / U.S. Attorney’s Office, the agency-average result should be interpreted as a dataset result rather than a broad cross-agency effectiveness benchmark.

## Project Objective

Identify measurable scam patterns and translate them into practical intervention opportunities for a fraud-risk organization.

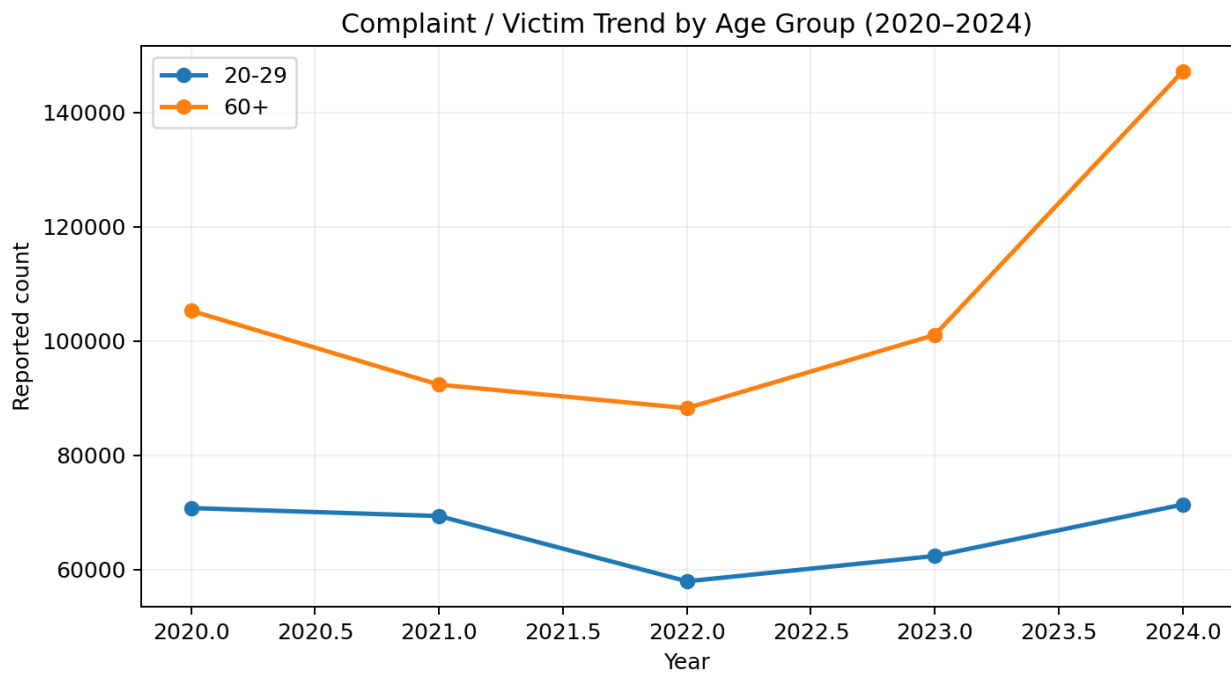
## Data & Methodology

The workbook contains structured source data and analysis sheets. Methods used include growth calculations, frequency analysis, splitting combined platform/payment fields, ranking, visualization, and government-action aggregation.

Important limitation: the age series uses “Victims” for 2020–2022 and “Complaints” for 2023–2024, so the five-year series is best described as a reported victim/complaint trend rather than a perfectly consistent complaint series. The narrative sample is limited.

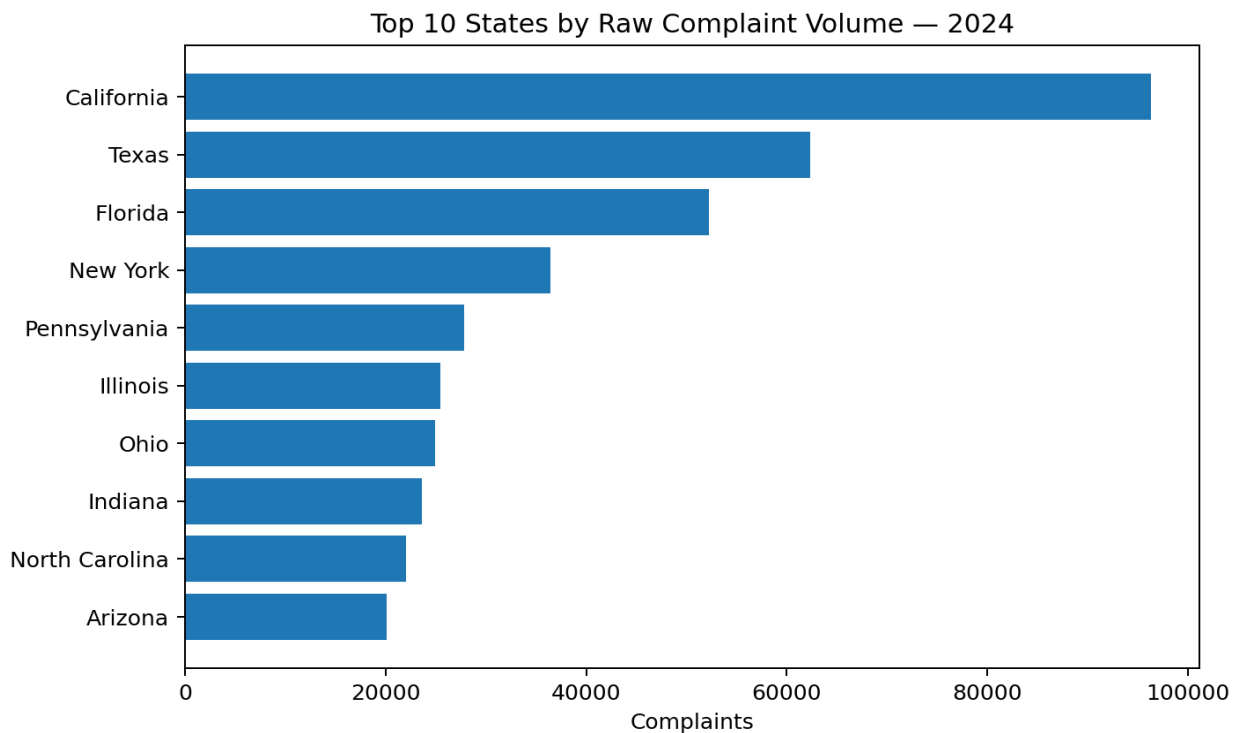
# 1.1 Victim Demographics & Geographical Hotspots

## Age-Group Trend



Observation: The 60+ group is the strongest growth signal in the supplied comparison. Its reported count increased by approximately 39.7% from 2020 to 2024, compared with approximately 0.9% for ages 20–29. This supports prioritizing older adults for targeted awareness and intervention.

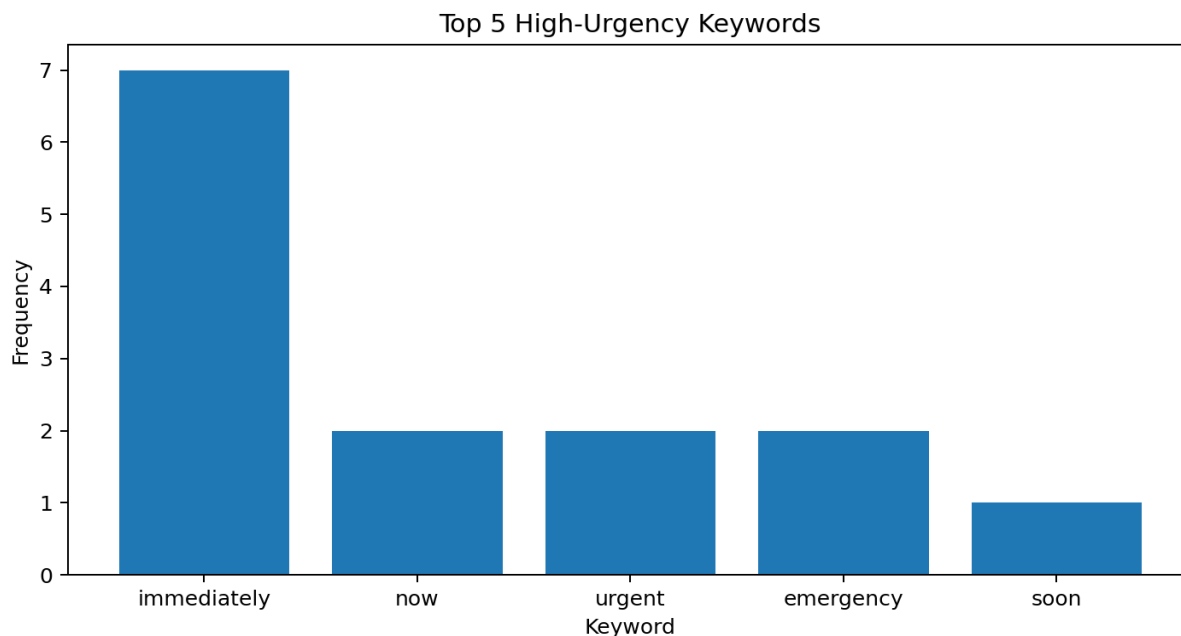
## 2024 Geographic Hotspots



California leads the supplied 2024 state records with 96,265 complaints, followed by Texas (62,347) and Florida (52,191). The ranking is based on raw complaint volume, as required by the project.

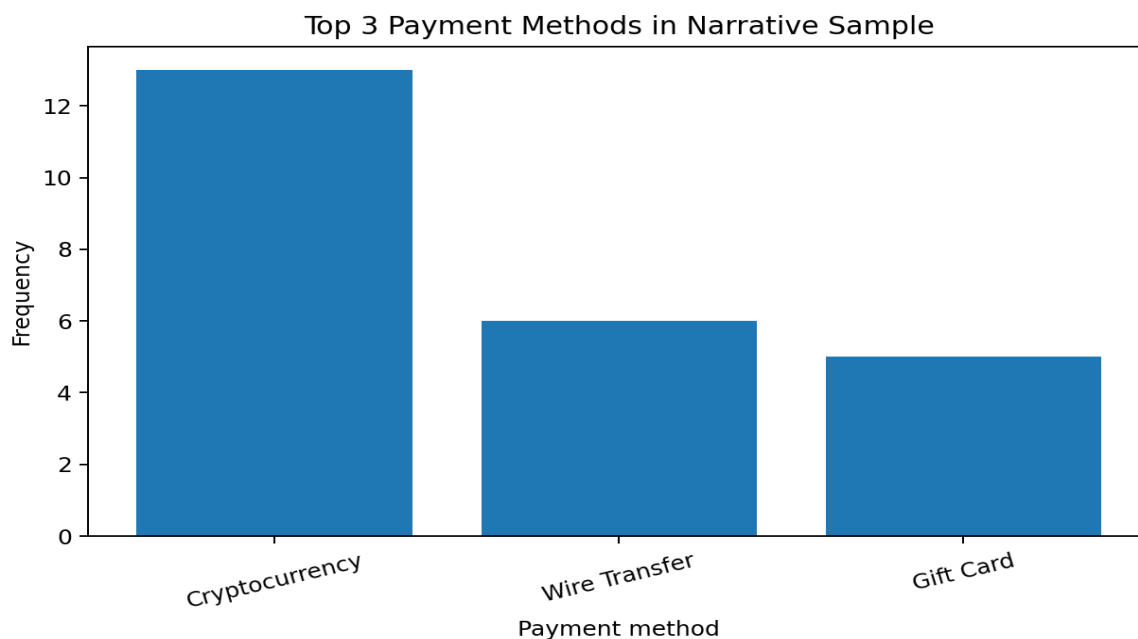
## 1.2 Scammer Tactics & Intervention Analysis

### Urgency Language



The five most frequent urgency keywords are immediately (7), now (2), urgent (2), emergency (2), and soon (1). This supports urgency-language detection as an early intervention signal.

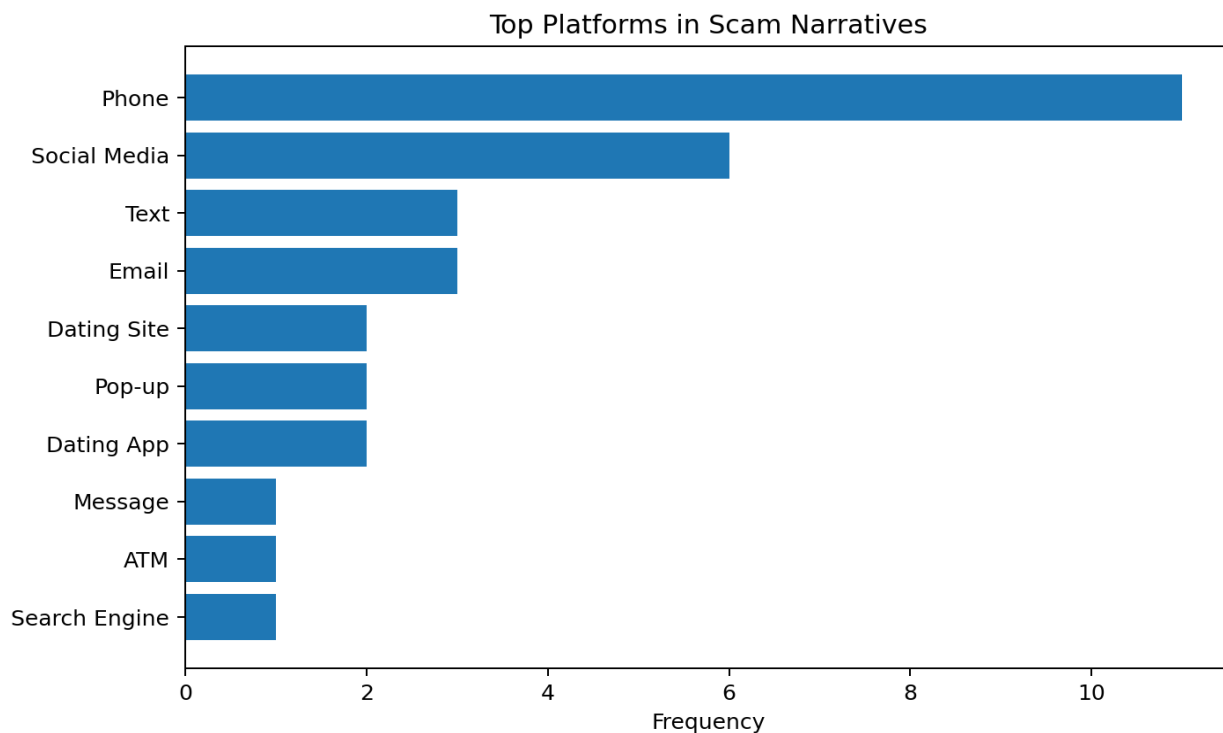
### Payment Methods



Cryptocurrency is the most frequently observed payment method in the collected narrative sample, followed by wire transfer and gift card.

## 1.2 Scammer Tactics — Platform & Script

### Platform Prevalence



Phone is the most frequent individual platform after combined descriptions are split, followed by social media. This indicates that intervention should not focus on a single channel.

### Common Imposter-Fraud Script

- Initial contact through phone, text, email, social media, or another platform.
- Impersonation of a trusted organization, person, support team, or authority.
- Urgency, fear, or a threat is introduced to reduce the victim's time for verification.
- A specific demand is made, such as payment or disclosure of information.
- The victim is pressured to act quickly and may be discouraged from seeking help.
- The scammer attempts to complete the transaction or obtain credentials before verification.

Intervention opportunity: the highest-value intervention point is before the victim provides credentials or completes a high-risk payment.

## 1.2 Payment-Method Shift: 2023 vs 2024

| Payment Method              | 2023 | 2024 |
|-----------------------------|------|------|
| Cash                        | 0    | 2    |
| Cryptocurrency              | 3    | 3    |
| Gift Card                   | 3    | 2    |
| Money Transfer              | 0    | 1    |
| Not primarily payment-based | 0    | 1    |
| Payment App                 | 1    | 0    |
| Precious Metals             | 0    | 2    |
| Wire Transfer               | 1    | 1    |

Observation: cryptocurrency remained at 3 mentions in both 2023 and 2024. Gift-card mentions declined from 3 to 2, while cash and precious metals appeared with 2 mentions each in 2024. Within this limited sample, the payment mix appears more diversified rather than shifting completely from one method to another. This is a sample-level observation, not a population-level industry trend.

### Business Implication

A defensive system should monitor multiple payment rails and update risk rules as scammers change their preferred collection methods.

# 1.3 Government Response & Future Defense

## Government Action Analysis

The structured government dataset contains 10 anti-scam actions with agency, targeted scam type, date, estimated funds recovered, action description and source information.

| Agency                       | Actions | Average estimated recovery |
|------------------------------|---------|----------------------------|
| DOJ / U.S. Attorney's Office | 10      | \$3,565,319                |

SQL-style question result: the highest average estimated recovery in the supplied dataset belongs to DOJ / U.S. Attorney's Office at approximately \$3,565,319 per action. Since all 10 supplied actions are represented under this agency grouping, the dataset does not provide a meaningful cross-agency comparison.

## Future of Anti-Scamming — 3 Technology Strategies

- 1. Real-time AI scam-language detection:** detect urgency, threats, impersonation cues and suspicious demands before a victim completes a risky action.
- 2. AI/deepfake and identity-risk detection:** analyze suspicious voice, text, image and identity patterns to flag convincing impersonation attempts.
- 3. Adaptive payment and threat intelligence:** connect scam narratives, payment indicators and emerging tactics so detection rules can continuously adapt as criminals change channels and payment methods.

# 1.3 Government Response — Strategic Gap & Recommendations

## Major Legislative / Enforcement Gap

A major operational gap is the speed and coordination mismatch between rapidly changing, cross-platform scam operations and fragmented detection/enforcement. A scam can move across phone, social media and financial or crypto channels before one organization sees the complete pattern.

Psyliq opportunity: build an early-warning threat-intelligence layer that combines scam-language, behavioral and payment-risk signals to identify coordinated scam patterns earlier and route high-risk events for verification or intervention.

## Key Recommendations

- Prioritize older adults with targeted awareness, trusted-contact alerts and simple verification prompts.
- Use urgency, threat and impersonation language as real-time intervention signals.
- Apply risk-based warnings before high-risk payment actions.
- Continuously update threat intelligence using new scam phrases, platforms, payment methods and public-sector intelligence.

## Conclusion

The strongest practical opportunity is intervention before a victim completes a high-risk action or payment. Combining demographic targeting, geographic concentration, narrative intelligence, platform signals and adaptive payment intelligence can give Psyliq a more proactive defensive capability.

## Source & Data Note

All project-specific figures and source references are based on the supplied Data\_Matrix\_P1.xlsx workbook. The workbook should remain the source-of-record for the report's underlying citations and compiled data.